

What Additive Risk-Adjusted Rewards Actually Optimise in Reinforcement Learning for Asset Allocation

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Research Proposal

Abstract

Reinforcement learning is now routinely applied to multi-period asset allocation with risk-adjusted rewards: Sharpe ratios, variance penalties, drawdown penalties. These rewards are summed across periods and optimised by algorithms whose correctness rests on the dynamic programming principle. Mean-variance objectives violate that principle. The optimisation is therefore performed on an additive *surrogate* rather than on the objective the reward was written to express, and the literature does not establish what the surrogate’s optimum is.

This project isolates that discrepancy as a distinct and previously unnamed source of error. Writing J for the true mean-variance objective, π^{pre} for the pre-committed optimum, π^{eq} for the equilibrium policy, π^r for the maximiser of the additive surrogate induced by a reward r , and $\hat{\pi}$ for the output of a learning algorithm, the total error decomposes exactly as

$$J(\pi^{\text{pre}}) - J(\hat{\pi}) = \underbrace{J(\pi^{\text{pre}}) - J(\pi^{\text{eq}})}_{\varepsilon_{\text{incons}}} + \underbrace{J(\pi^{\text{eq}}) - J(\pi^r)}_{\varepsilon_{\text{surr}}} + \underbrace{J(\pi^r) - J(\hat{\pi})}_{\varepsilon_{\text{learn}}}.$$

The first term is the known price of time-consistency; the third is the optimisation error addressed by existing convergence theory. The middle term, which I call the *surrogate gap*, is the contribution of reward design itself, and is invisible to every backtest-based evaluation in the applied literature. The primary theoretical target is a bound of the form $\varepsilon_{\text{surr}}(r) \leq C \Lambda(r)^\gamma$ together with a characterisation of the rewards for which $\Lambda(r) = 0$. The project is theoretical; numerical work uses a synthetic market in which the pre-committed optimum is available in closed form, so that $\varepsilon_{\text{surr}}$ can be measured against ground truth rather than inferred from a backtest.

1 Motivation

Reinforcement learning has become a standard tool for multi-period asset allocation. The survey of Bai et al. (2024) catalogues roughly twenty portfolio-management papers; the survey of Hambly et al. (2023) covers the same ground from the mathematical finance side. Across both, one pattern is consistent. Practitioners want a policy that trades return against risk, so they encode risk into the reward—a Sharpe ratio, a differential Sharpe ratio, a variance penalty—and hand the resulting sum of per-period rewards to a standard solver: a deterministic policy gradient method, an actor–critic method, or PPO.

There is a mathematical problem with this construction that the applied literature does not address. These algorithms are derived from the dynamic programming principle and are correct because the Bellman equation holds. For mean-variance objectives it does not. Hambly, Xu and Yang state this directly in their discussion of portfolio optimisation: the mean-variance framework suffers from time-inconsistency, in that a strategy chosen as optimal at time t is no longer optimal at a later time s , and the Bellman equation fails.

So when a practitioner trains PPO on a Sharpe-based reward and reports an improved backtest, it is not clear which object has been computed. It is not the pre-committed mean-variance optimum, which requires the embedding of Li and Ng (2000) and cannot be reached by dynamic programming on the original objective. It is not obviously the equilibrium policy of consistent planning either. Nor does the convergence theory settle the question: Xiong et

al. (2022) establish that deterministic policy gradient converges to a stationary policy, but a stationary point of an additive surrogate is not a solution to the problem the reward was written to express.

The purpose of this project is to name that discrepancy, bound it, and measure it.

2 Mathematical setting

2.1 Multi-period mean-variance allocation

The mean-variance criterion of Markowitz is a single-period model; its multi-period extension is the setting here. Following the formulation in Hambly et al. (2023), which follows Li and Ng (2000), let there be m risky assets and horizon T . The random rates of return at time t are

$$e_t = (e_t^1, \dots, e_t^m)^\top, \quad \mathbb{E}[e_t] = \mu_t, \quad \text{Cov}(e_t) = \Sigma_t, \quad (1)$$

assumed independent across t . A portfolio is a weight vector on the unit simplex,

$$\Delta^{m-1} = \left\{ w \in \mathbb{R}^m : w_i \geq 0 \ \forall i, \ \sum_{i=1}^m w_i = 1 \right\}, \quad (2)$$

and, writing u_t for the vector of amounts invested and x_t for wealth, the wealth process evolves as

$$x_{t+1} = \sum_{i=1}^{m-1} e_t^i u_t^i + \left(x_t - \sum_{i=1}^{m-1} u_t^i \right) e_t^m, \quad t = 0, \dots, T-1. \quad (3)$$

The investor's objective, evaluated from the initial state, is

$$J(u) = \mathbb{E}[x_T] - \varphi \text{Var}(x_T), \quad \varphi > 0. \quad (4)$$

Li and Ng solved (4) by embedding it in an auxiliary linear-quadratic problem. The resulting pre-committed optimal control is affine in wealth,

$$u_t^*(x_t) = \alpha_t x_t + \beta_t, \quad (5)$$

with α_t, β_t explicit functions of μ_t and Σ_t . Equation (5) is central to this proposal: in a market whose moments are known, the exact optimal policy is available in closed form, so any learned policy can be measured against it directly rather than against a backtest statistic.

2.2 The MDP formulation and the failure of dynamic programming

A finite-horizon MDP is a tuple (S, A, P, r) . Following Hambly et al. (2023) and the standard treatment in Sutton and Barto (2018), the value function is

$$V_t^*(s) = \sup_{\Pi} \mathbb{E}^\Pi \left[\sum_{u=t}^{T-1} r_u(s_u, a_u) + r_T(s_T) \mid s_t = s \right], \quad (6)$$

and the dynamic programming principle gives the Bellman equation

$$V_t^*(s) = \sup_{a \in A} \left\{ \mathbb{E}[r_t(s, a)] + \mathbb{E}_{s' \sim P_t(s, a)} [V_{t+1}^*(s')] \right\}. \quad (7)$$

Every algorithm considered here— Q -learning, actor-critic, deterministic policy gradient (Silver et al., 2014), its deep variant DDPG, and PPO (Schulman et al., 2017)—is a stochastic approximation to (7) or to the policy-gradient identity that accompanies it.

Now evaluate the mean-variance objective from an arbitrary state,

$$J(t, x, u) = \mathbb{E}_{t,x}[x_T] - \varphi \left(\mathbb{E}_{t,x}[x_T^2] - (\mathbb{E}_{t,x}[x_T])^2 \right). \quad (8)$$

The tower property applies to each expectation separately, but $(\mathbb{E}_{t,x}[x_T])^2$ is a nonlinear functional of a conditional expectation and the square of an expectation does not itself satisfy the tower property. Consequently

$$J(t, x, u^*) \neq \sup_a \mathbb{E}[J(t+1, x_{t+1}, u^*)] \quad \text{in general,} \quad (9)$$

which is the formal content of the statement that the Bellman equation does not hold. Time-inconsistency in this sense goes back to Strotz (1955); the modern treatment of Markovian time-inconsistent control is due to Björk and Murgoci (2014).

2.3 Three inequivalent notions of optimality

Once (7) fails, the word *optimal* is ambiguous. Three distinct objects are studied, and they do not coincide.

Pre-committed.

The solution of (4) computed once at time 0 and executed without revision. This is the Li–Ng policy (5). It is optimal from the standpoint of time 0 only.

Equilibrium (consistent planning).

A policy from which no single-period deviation is profitable, given that all later selves continue with the same policy. Formally, π^{eq} is an equilibrium if for all t, x and all admissible a ,

$$J(t, x, \pi^{\text{eq}}) \geq J(t, x, (a, \pi_{t+1:T}^{\text{eq}})). \quad (10)$$

Myopic.

The single-period optimiser applied repeatedly, ignoring the horizon.

Basak and Chabakauri (2010) characterise the equilibrium mean-variance policy and show it differs from the pre-committed one. Neither is reachable by naive dynamic programming on (4).

3 Previous work and its limitations

3.1 Applied reinforcement learning for portfolio management

Bai et al. (2024) classify the rewards used across the papers they survey. Alongside plain return and profit-and-loss, one of their principal categories is volatility-adjusted return, defined to include the Sharpe ratio and its variants. Hambly et al. (2023) report the same for portfolio optimisation: the reward signals in use are portfolio return, the differential Sharpe ratio, and profit. Two of the works they cite compare reward choices directly and find that the choice matters for stability, with one concluding that Q -learning is more sensitive to the choice of value function than the policy-based alternative, and another reporting that DDPG and PPO fail to recover the optimal policy even on the training set. None of this work analyses what the additive optimisation targets.

3.2 Time-inconsistent stochastic control

Strotz (1955) identified the phenomenon; Björk and Murgoci (2014) give the general Markovian theory in discrete time; Basak and Chabakauri (2010) characterise the equilibrium mean-variance policy; Li and Ng (2000) give the pre-committed solution in closed form. This literature assumes exact optimisation. It does not consider what happens when the policy is obtained by a learning algorithm operating on an additive reward.

3.3 Convergence theory for policy-gradient methods

Hambly et al. (2023) record that no convergence guarantee had been established for DPG or DDPG at the time of writing. Since then Xiong et al. (2022) have analysed single-timescale DPG in both on-policy and off-policy settings and established convergence to an ε -accurate stationary policy with sample complexity $\mathcal{O}(\varepsilon^{-2})$, together with a rate under the Gaussian exploration noise used in practice. These results bound the distance between the learned policy and a stationary point *of the objective actually being optimised*. They say nothing about the relation between that objective and (4).

3.4 Continuous-time treatments

Wang and Zhou (2020) study the entropy-regularised continuous-time mean-variance problem and prove that the optimal exploratory policy is Gaussian with decaying variance. This is a correct treatment of exploration for the mean-variance criterion, but in continuous time and for the exploratory reformulation; it does not describe the discrete-time additive-reward practice that dominates the applied literature.

3.5 Summary of the gap

Research direction	What it provides	Remaining limitation
Applied RL for portfolio management	Empirical methods, reward taxonomy, extensive backtest evidence	No analysis of what the additive surrogate optimises
Time-inconsistent control	Pre-committed and equilibrium solutions; closed form for mean-variance	Assumes exact optimisation; no learning error
Convergence theory for policy gradient	Convergence to stationary points of the optimised objective	Silent on the relation between that objective and (4)
Continuous-time exploratory control	Correct treatment of exploration for mean-variance	Continuous time; does not cover discrete-time additive rewards

Table 1: Relationship between existing approaches and the proposed research gap.

The gap is the absence of any analysis connecting three mechanisms that are individually well understood:

$$\underbrace{\text{time-inconsistency}}_{\text{objective structure}} + \underbrace{\text{additive reward surrogacy}}_{\text{reward design}} + \underbrace{\text{learning error}}_{\text{algorithm}}$$

in multi-period asset allocation. The applied literature supplies the third and ignores the first two; the control literature supplies the first and ignores the third; nobody treats the second at all.

4 Central research question

When a risk-adjusted reward is optimised additively by a reinforcement learning algorithm, how far is the resulting policy from the mean-variance optimum, and how does that distance depend on the reward?

Fix a per-period reward r and write $J_r(\pi) = \mathbb{E}[\sum_t r_t]$ for the additive objective it induces, $\pi^r = \arg \max_{\pi} J_r(\pi)$ for its maximiser, and $\hat{\pi}$ for the policy returned by a learning algorithm

applied to J_r . Since J is a scalar functional of a policy, the total shortfall admits the exact decomposition

$$J(\pi^{\text{pre}}) - J(\hat{\pi}) = \underbrace{[J(\pi^{\text{pre}}) - J(\pi^{\text{eq}})]}_{\varepsilon_{\text{incons}}(\varphi)} + \underbrace{[J(\pi^{\text{eq}}) - J(\pi^r)]}_{\varepsilon_{\text{surr}}(r)} + \underbrace{[J(\pi^r) - J(\hat{\pi})]}_{\varepsilon_{\text{learn}}}. \quad (11)$$

The three terms have different status. The first is the price of time-consistency, an irreducible quantity determined by φ and the market, characterised by Basak and Chabakauri (2010). The third is optimisation and estimation error, the object of the convergence theory discussed above. The middle term is the cost of encoding a risk-adjusted objective as a sum of per-period rewards. It is not addressed anywhere in the literature, and it is invisible to backtest-based evaluation because $J(\pi^{\text{pre}})$ is unknown on historical data.

Target theorem

The primary theoretical goal is a bound on the middle term, with the constants and exponent to be determined.

Theorem 1 (Target). *Under regularity assumptions on the return distribution and the admissible policy class, there exist a functional Λ on the reward class $\mathcal{R}$, a constant $C > 0$ and an exponent $\gamma > 0$ such that for every $r \in \mathcal{R}$,*

$$0 \leq \varepsilon_{\text{surr}}(r) \leq C \Lambda(r)^\gamma,$$

with $\varepsilon_{\text{surr}}(r) = 0$ if and only if $\Lambda(r) = 0$. Moreover $\Lambda(r) = 0$ for the logarithmic return reward, and $\Lambda(r) > 0$ for the variance-penalised and Sharpe-based rewards in common use.

Combining Theorem 1 with (11) and an existing bound on $\varepsilon_{\text{learn}}$ yields the end-to-end statement

$$J(\pi^{\text{pre}}) - J(\hat{\pi}) \leq \varepsilon_{\text{incons}}(\varphi) + C \Lambda(r)^\gamma + \varepsilon_{\text{learn}}, \quad (12)$$

which separates the three sources of shortfall and attributes each to its cause: the choice of risk criterion, the choice of reward, and the choice of algorithm.

The functional Λ is the object to be constructed. Informally it should measure the failure of r to induce a time-consistent objective; the natural candidate is the magnitude of the violation of (7) generated by r , but establishing the right form is part of the work.

5 Proposed method

The method has three analytical steps and one computational one. Step 1 makes the decomposition (11) rigorous and constructs the functional Λ that Theorem 1 requires. Step 2 treats the one reward in common use whose formulation is not even Markovian on the state space it is usually given, and shows what augmentation repairs it. Step 3 measures the resulting quantities against a market in which the true optimum is known in closed form, which is the step that makes the question answerable at all. Step 4 is implementation. Steps 1–3 are independent of the computational work and of each other in the sense set out in Section 8.

5.1 Step 1: exactness of the decomposition and construction of Λ

The decomposition (11) is a telescoping identity and requires only that each policy be admissible and J well defined; establishing this rigorously is the first step. The substantive work is the construction of Λ and the proof of Theorem 1.

Two limiting cases are already clear and serve as consistency checks. With logarithmic return and no risk term the objective becomes expected log-wealth, which is additive across

periods and therefore time-consistent by construction; this is the Kelly criterion and dynamic programming is valid, so $\Lambda = 0$ and $\varepsilon_{\text{surr}} = 0$. This explains why the log-return reward used by Jiang et al. (2017) and Li (2019) rests on firmer ground than the field’s typical Sharpe-based rewards, a point neither paper makes. At the other extreme, a terminal-only variance penalty fails additivity outright and the analysis reduces to the known time-inconsistency of (4).

If the general characterisation proves harder than expected, a sharp counterexample in a two-asset, two-period market—exhibiting an explicit reward r , an explicit gap $\varepsilon_{\text{surr}}(r) > 0$, and the resulting misallocation—is achievable independently and constitutes a result on its own.

5.2 Step 2: restoring the Bellman equation by state augmentation

The differential Sharpe ratio of Moody and Saffell (2001), reported by Hambly et al. (2023) as one of the standard reward choices in this literature, maintains exponentially weighted moment estimates with decay η :

$$A_t = A_{t-1} + \eta(\rho_t - A_{t-1}), \quad B_t = B_{t-1} + \eta(\rho_t^2 - B_{t-1}), \quad (13)$$

$$r_t^{\text{DSR}} = \frac{B_{t-1} \Delta A_t - \frac{1}{2} A_{t-1} \Delta B_t}{(B_{t-1} - A_{t-1}^2)^{3/2}}. \quad (14)$$

Since r_t^{DSR} depends on (A_{t-1}, B_{t-1}) , it is not a function of the market state alone and the MDP of Section 2.2 is misspecified as usually written. I propose the augmented state

$$z_t = (s_t, A_t, B_t) \in S \times \mathbb{R} \times \mathbb{R}_+ \quad (15)$$

and will verify that the augmented process is Markov, that (7) holds on the augmented space, and that the resulting fixed point is well defined. The objective is then to compute $\Lambda(r^{\text{DSR}})$ explicitly.

This yields a usable result in either direction. Either practitioners using the differential Sharpe ratio have been implicitly solving a well-posed augmented problem, in which case the augmentation should be made explicit and the state should include the moment estimates; or they have not, in which case the reward is being optimised over a state space that cannot support it.

5.3 Step 3: measurement against a closed-form optimum

This is where the project departs most sharply from the applied literature. Rather than backtesting, I work in a synthetic market whose moments are known, so that π^{pre} is available analytically from (5) and π^{eq} is computable by backward recursion.

For each candidate reward, a policy-gradient agent is trained and the policy-space distances

$$d_{\text{pre}}(\theta) = \left(\sum_t \mathbb{E} \|\pi_\theta(z_t) - u_t^*(x_t)\|^2 \right)^{1/2}, \quad d_{\text{eq}}(\theta) = \left(\sum_t \mathbb{E} \|\pi_\theta(z_t) - \pi^{\text{eq}}(x_t)\|^2 \right)^{1/2} \quad (16)$$

are reported, together with an empirical estimate $\widehat{\varepsilon}_{\text{surr}}$ obtained by evaluating J at the trained and reference policies. The comparison of interest is whether $\widehat{\varepsilon}_{\text{surr}}$ tracks the bound of Theorem 1, and whether d_{pre} or d_{eq} is smaller for each reward. Because (5) is exact, these are measurements against ground truth rather than against a benchmark strategy.

All quantities are reported as means and spreads over at least five random seeds, since Li (2019) documents that agents differing only in seed can behave very differently and no conclusion drawn from a single run would be safe.

5.4 Step 4: numerical implementation

The primary contribution of this project is theoretical. This step serves to validate the predicted behaviour of $\varepsilon_{\text{surr}}$ and is not itself the contribution.

The synthetic market is deliberately small: a handful of assets with specified μ and Σ , chosen so that (5) is exact and the equilibrium recursion is tractable. The learning algorithm is taken from a standard implementation rather than written from scratch, since the algorithm is not the object of study and using a standard implementation removes one source of unexplained variation. Candidate rewards are the logarithmic return, the return net of a linear transaction cost, the variance-penalised return, and the differential Sharpe ratio with the augmentation of Step 2.

A final step re-runs the trained policies on historical data, purely to establish whether the discrepancies identified in the controlled setting are visible at all in the backtest statistics the field normally reports. I expect them not to be, which is the empirical content of the third row of the gap table.

6 Expected contributions

C1. Error decomposition.

Isolation of the surrogate gap $\varepsilon_{\text{surr}}$ as a distinct source of error in reinforcement learning for risk-adjusted objectives, via the exact decomposition (11).

C2. Characterisation.

A bound $\varepsilon_{\text{surr}}(r) \leq C\Lambda(r)^\gamma$ together with necessary and sufficient conditions for $\varepsilon_{\text{surr}}(r) = 0$, and the resulting end-to-end statement (12).

C3. Augmented formulation.

A Markov formulation of the differential Sharpe ratio reward under which (7) is restored, with Λ computed for it.

C4. Measurement protocol.

A procedure for measuring the surrogate gap against a closed-form optimum, replacing backtest comparison for this class of question.

C5. Numerical validation.

Empirical confirmation, in a controlled market, of the predicted dependence of $\varepsilon_{\text{surr}}$ on the reward.

7 Research plan

The phases are sequenced so that each yields something usable on its own. Phase 3 is self-contained and could be reordered before Phase 2 if the construction of Λ proves slow. Phase 4 requires the reference policies from Phase 1 and the augmentation from Phase 3.

Working alone, I expect the analytical phases to dominate the schedule, and the scoping reflects that: the synthetic market is low-dimensional and the algorithm is taken off the shelf.

8 Risks and scope

The general characterisation in Theorem 1 may prove intractable under fully general assumptions on the reward class. The project is therefore staged. The decomposition (11) is elementary and secure. A counterexample establishing $\varepsilon_{\text{surr}} > 0$ for a reward in current use, with an explicit magnitude, is achievable independently of the general theorem and would already be

Phase	Main objectives
Phase 1	Formalise the discrete-time mean-variance MDP. Define π^{pre} , π^{eq} , π^r precisely and establish that (11) is exact under stated admissibility conditions.
Phase 2	Construct Λ . Prove Theorem 1, or failing that produce the two-asset counterexample with an explicit $\varepsilon_{\text{surr}} > 0$. Verify the Kelly case $\Lambda = 0$.
Phase 3	Augmented MDP for the differential Sharpe ratio: Markov property, validity of (7) on the augmented space, computation of $\Lambda(r^{\text{DSR}})$.
Phase 4	Implement the synthetic market, the Li–Ng reference policy, and the equilibrium backward recursion. Train agents on each candidate reward across seeds.
Phase 5	Measure $\widehat{\varepsilon_{\text{surr}}}$ against the predicted bound, run the historical comparison, finalise proofs and write up.

Table 2: Staged research plan. Phases 1–3 are analytical and independent of the computational work.

a contribution: it would demonstrate that a widely used reward design incurs a cost that the field’s evaluation methodology cannot detect. The augmented formulation of Step 2 is likewise independent. Only the strongest form—a sharp bound with explicit exponent across a general reward class—depends on everything else succeeding.

Two limitations of scope should be stated. The synthetic market assumes returns independent across periods with known first two moments, matching the setting in which (5) is derived; real returns satisfy neither condition, and results obtained there transfer to real markets as statements about the algorithms rather than about achievable performance. This is a deliberate trade: external validity is given up in order to obtain a ground truth against which the central question can be settled at all, and the absence of such a ground truth is precisely what has left the question open. Second, the analysis is restricted to the mean-variance family and its reward-based surrogates; conditional value-at-risk and the risk-sensitive formulations listed among the future directions of Bai et al. (2024) raise related but distinct dynamic-consistency issues and fall outside this project.

9 Conclusion

The applied literature optimises risk-adjusted rewards additively, using algorithms justified by a principle that the underlying objective violates, and evaluates the result with a statistic that cannot detect the discrepancy. This proposal names the discrepancy, places it inside an exact error decomposition alongside the two error sources that are already understood, and proposes to bound it and measure it against a case where the true optimum is known.

The desired outcome is a statement of the form

$$J(\pi^{\text{pre}}) - J(\widehat{\pi}) \leq \varepsilon_{\text{incons}}(\varphi) + C\Lambda(r)^\gamma + \varepsilon_{\text{learn}},$$

separating the price of time-consistency, the cost of reward design, and the error of the learning algorithm. Whichever way the constants fall, the result identifies what an entire applied literature has actually been computing.

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